

Full Featured IEEE Two-Column Paper Template

First Author, Member, IEEE, Second Author, Fellow, IEEE

Abstract—This template contains practical working examples for single/two-column equations, TikZ flowcharts, multi-column tables, matrix arrays, subfigure insertion, and automated BibTeX reference management inside an IEEE journal format.

Index Terms—IEEEtran, TikZ Flowchart, Subfigures, Equations, Matrix Arrays, BibTeX.

I. INTRODUCTION

THIS document demonstrates how to use graphics, arrays, flowcharts, and structured tables inside the IEEE LaTeX document class. Modern communication theory was laid out by Shannon [1], while modern deep learning architectures build upon transformer models [2] and foundational neural network design [3]. Guidelines regarding general manuscript structure follow official IEEE formatting directives [4].

II. EQUATIONS AND MATRIX ARRAYS

Standard aligned multi-line equation:

$$\begin{aligned} f(x) &= \int_0^\infty e^{-x^2} dx + \sum_{k=1}^n \frac{1}{k^2} \\ &= \frac{\sqrt{\pi}}{2} + \zeta(2). \end{aligned} \quad (1)$$

A 3×3 matrix array representation:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}. \quad (2)$$

III. TIKZ FLOWCHART CREATION

Fig. 1 shows a native TikZ block flowchart designed to fit cleanly inside a single column.

IV. IMAGE INSERTION

To insert standard figures or subfigures, use the setups below.

V. TABLE DESIGN

Table I demonstrates an IEEE-compliant table structure utilizing ‘booktabs’ styling.

VI. CONCLUSION

This preamble provides all dependencies required for complex IEEE technical papers, with full citation support linking directly to ‘references.bib’.

F. Author and S. Author are with the Department of Electrical Engineering, University Name, City, Country (e-mail: author@university.edu).

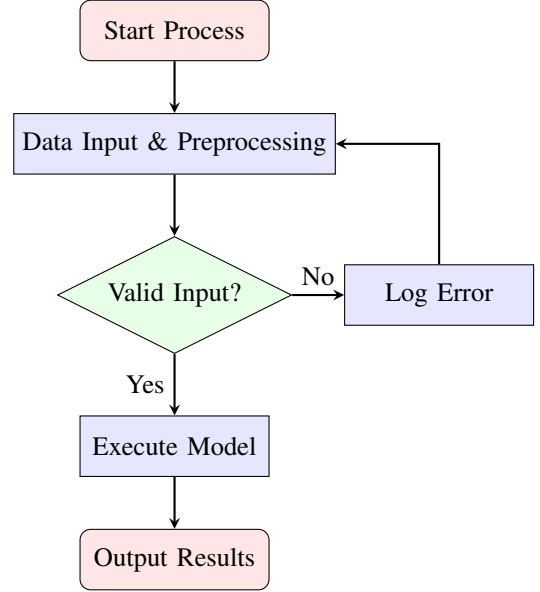


Fig. 1. System architecture and operational flowchart.

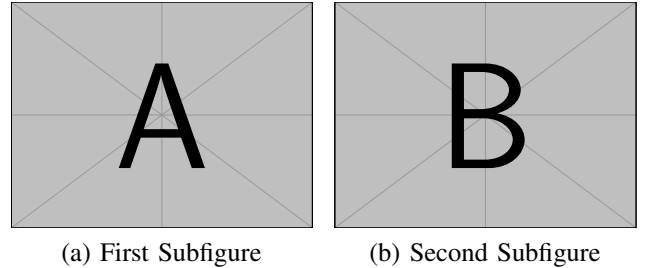


Fig. 2. Example of two subfigures side by side within one column.

TABLE I
PERFORMANCE COMPARISON ACROSS BASELINE MODELS

Method	Accuracy (%)	Latency (ms)	Parameters
Model A	88.4	12.5	1.2M
Model B	91.2	18.0	3.4M
Proposed	95.6	10.2	0.9M

REFERENCES

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